



Divya Sailaja Kumari Gurram

Data Engineer

+18136475642 sailaja.gurram61@gmail.com



More than 4+ years of IT experience in Data Warehousing and Business Intelligence. Experience in analysing requirements, designing, implementing, and unit testing various Data Warehousing projects.

- SUMMARY**
- 4+ years of expertise designing, developing, and executing data pipelines and data lake requirements in numerous companies using the **Big Data Technology stack, Python, PL/SQL, SQL, REST APIs**, and the **Azure cloud platform**.
 - Expertise in building batch and real-time data processing systems leveraging **Azure services (ADLS, ADF, Databricks, Synapse Analytics)**.
 - Implemented **Big Data solutions** using **Hadoop** technology stack, including **PySpark, Hive, Sqoop, Avro** and **Thrift**. Develop **DDL** and **DML** statements for data modelling, data storage and looked into performance fine tuning. Experience in providing highly available and fault tolerant applications utilizing orchestration technologies like **Kubernetes** on **Google Cloud Platform**.
 - Expert in creating various **Kafka** producers and consumers for seamless data streaming with **Azure** services. Hands-on experience working with **DevOps** tools such as **Jenkins, Docker, Kubernetes, GoCD**, and **Autosys scheduler**.
 - Good understanding of **Spark Architecture** with **Databricks**, Structured Streaming. Setting **Microsoft Azure** with **Databricks**, **Databricks** Workspace for Business **Analytics**, Managing **Clusters** in **Databricks**, Managing the **Machine Learning Lifecycle**.
 - Highly skilled in using visualization tools like **Tableau, matplotlib, ggplot2** for creating **dashboards**.
 - Experience in **Windows Azure Services** like **PaaS, IaaS** and worked on storages like **Blob** (Page and **Block**), **SQL Azure**. Well experienced in deployment & configuration management and Virtualization.
 - Designed and developed **ETL** jobs for large scale data ingestion, transformation, cleansing, and migration between a variety of sources and sinks using **PySpark, Scala, SQL, Kafka, Airflow**.
 - Experienced in building **Snow Pipes**, migrating **Teradata** objects into **Snowflake** environment.
 - Hands-on experience modeling data (**Snowflake, Star Schema**) and developing pipelines for BI analytics and visualization using **Power BI, Tableau**.
 - Proficient in building **CI/CD** pipelines in **Jenkins** using pipeline syntax and **groovy** libraries.
 - Experience in **Cisco Cloud Center** to more securely deploy and manage applications in multiple data center, private cloud, and public cloud environments.
 - Designed and implemented a scalable data architecture on **Azure** using **Kubernetes, Terraform**, and **Snowflake**, enabling seamless data integration and processing across multiple data sources.

SKILLS

Languages	Python (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn), Java, C#, C++, R, SQL, Hive, Spark, PowerShell
Statistical Methods	Hypothesis Testing, ANOVA, Principal Component Analysis (PCA), TimeSeries, Correlation Multivariate Analysis.
Cloud Platform	Azure, Aws,GCP (Google Cloud Console, Google Cloud Shell)
Data Visualization	Tableau, Python (Matplotlib, Seaborn), R (ggplot2), PowerBI, Qlik View, D3.js., Kibana
Hadoop Ecosystem	Hadoop, Spark, Snowflake, MapReduce, HiveQL, HDFS, Sqoop, PigLatin
Data Sources	PostgreSQL, Microsoft SQL Server, MongoDB, MySQL, NoSQL, Oracle, PL/SQL, DynamoDB, HBase, Amazon Redshift, Teradata, RDBMS, Snowflake
Methodologies	Agile, Scrum, Kanban, Waterfall
Operating Systems	UNIX Shell Scripting (via PuTTYclient), Linux, Windows, MacOS.
Other Tools and Technologies	TensorFlow, Terraform, IaaS, PaaS, SaaS, Kera's, NLTK, SpaCy, Genism, Microsoft Office Suite, GitHub, MATLAB, Pega, SSIS, Rest, Jenkins, SSRS, DBT
Version Control Software	Git, TFS, CVS, SVN

PROFESSIONAL EXPERIENCE

Client: Bank United, Miami Lakes, FL, US

April 2024 - Present

Role: Azure Data Engineer

Description: BankUnited is a regional bank offering personalized financial services, including personal and business banking, mortgage solutions, and wealth management. Designed data integration solutions with Azure Data Factory and Databricks, optimizing ETL workflows and enhancing real-time processing with Spark and Kafka.

Responsibilities:

- Designing and implementing data integration solutions using **Azure Data Factory** to move data between various data sources, including on-premises and cloud-based systems.
- Applied ETL best practices by designing and implementing **ETL** workflows using **Talend** and SQL Server Integration Services (**SSIS**), which enhanced data processing efficiency
- Developed efficient data extraction, transformation, and loading (**ETL**) processes, ensuring optimal performance and reliability, which reduced data processing time
- Integrated **HDFS** with real-time data processing frameworks like **Spark** and **Kafka** to optimize data pipelines
- Wrote **oozie** scripts and setting up workflow using **Apache Oozie** workflow engine for managing and scheduling **Hadoop** jobs.
- Implemented **Azure data lake**, **Azure Data Factory** and **Azure databricks** to move and conform the data from on-premises to cloud to serve the analytical needs of the organization.
- Designed scalable **NoSQL** data solutions using **DynamoDB** for high-volume banking transactions, ensuring low-latency data retrieval and resilience against system failures
- Developed **data pipelines** and workflows in **Azure Databricks** to process and transform large volumes of data using **Python**, **Scala**, and **SQL**.
- Imported real-time weblogs using **Kafka**, ingesting data into **Spark** Streaming for quality checks and business logic transformations.
- Supported continuous storage in **Azure Data Lake Storage (ADLS)** with snapshot configurations.
- Designed and built APIs in **Snowflake** and **Azure Data Factory (ADF)** using **REST** and **RPC** protocols to enable seamless **data integration** and retrieval, enhancing data accessibility and supporting analytical workflows.
- Developed and deployed **Azure Functions** to create serverless, event-driven workflows, enabling seamless integration with Azure services and ensuring efficient data processing and automation of key tasks.
- Created Data tables utilizing **PyQt** to display customer and policy information and add, delete, update customer records.
- Ensured data quality and accuracy with custom **SQL** and **Hive scripts** and created data visualizations using **Python** and **Tableau** for improved insights and decision-making.
- Monitored servers using **Nagios** and **CloudWatch**, leveraging the **ELK Stack (Elasticsearch and Kibana)** for indexing and visualizing real-time analytics. Actively participated in the creation and maintenance of **CI/CD** pipelines using **Jenkins**, enhancing deployment efficiency.
- Developed database triggers and **stored procedures** using **T-SQL**, ensuring efficient data handling and retrieval processes. Utilized **DBT** (Data Build Tool) for transformation workflows, automating the creation and management of data models. Conducted query optimization and performance tuning in **Snowflake**, resulting in improved query response times and reduced costs.
- Managed data migration projects, including data import/export to **MongoDB**, ensuring data integrity and consistency. Utilized **Jira** for ticketing and issue tracking throughout the **Software Development Life Cycle (SDLC)**
- Designed and implemented data warehousing solutions using **Azure Synapse Analytics**, including building data models, **ETL** processes, and analytical queries.

Environment: Apache, APII, Azure, Azure Synapse Analytics, CI/CD, Data Factory, Docker, Elasticsearch, ETL, Factory, Hive, IaaS, Java, Jenkins, Jira, Kafka, Kubernetes, lake, Oozie, PaaS, Python, Scala, Snowflake, Spark, Spark Streaming, SQL, Tableau, NoSQL, DynamoDB, HDFS

Client: Advent Health, Orlando, FL, USA

July 2023 – March 2024

Role: Data Engineer

Description: Advent Health Orlando is a premier healthcare provider offering comprehensive medical services, with a focus on compassionate care and improving patient well-being. Optimized data pipelines and real-time processing with AWS and Azure, ensuring secure healthcare data management, compliance, and automation.

Responsibilities:

- Loaded data from **BDW** Oracle database and Teradata into **HDFS** using **Sqoop**, and implemented **AJAX**, **JSON**, and **JavaScript** to create interactive web screens, enhancing user experience. Executed **ETL** operations to extract data from source systems and load it into Azure Data Storage services, improving data accessibility.
- Set up and worked on **Kerberos** authentication principals to establish secure network communication on cluster and testing of **HDFS**, **Hive**, **Pig** and **MapReduce** to access cluster for new users.
- Managed and optimized backend systems for Electronic Medical Records (**EMR**), ensuring secure data storage, retrieval, and processing.
- Designed and implemented scalable data pipelines to integrate clinical and administrative data, enhancing system efficiency and compliance with healthcare standards (**HIPAA**).
- Designed and implemented data pipelines utilizing **AWS Glue** and **S3** for efficient data extraction, transformation, and loading into data lakes.
- Configured and optimized **AWS Redshift** clusters to manage large-scale analytical queries, ensuring performance efficiency.
- Developed real-time data ingestion and processing workflows using **AWS Kinesis** and **FireHose**, enabling near-real-time analytics.
- Developed data-driven healthcare solutions using **Python**, automating data extraction, transformation, and validation processes for clinical and operational datasets, improving data quality and reducing processing time by 30% for regulatory reporting and analytics.
- Imported real-time weblogs using **Kafka** as a messaging system and ingested the data to **Spark Streaming**, performing data quality checks and categorizing data with bad and passable flags, which enhanced data reliability for downstream applications.
- Supported continuous storage in **ADLS** and configured snapshots, wrote entities in Spark with named queries to enhance database interaction.
- Developed **PySpark/Spark SQL** scripts to analyze customer behaviors, creating data tables with **PyQt** to manage customer records, which improved data accessibility for the team.
- Utilized **Tableau** for connecting to various data sources and building graphs, contributing to the design, development, deployment, testing, implementation, and support of projects.
- Implemented **NoSQL** solutions using **MongoDB** for healthcare data management, optimizing the storage and retrieval of unstructured clinical data and patient records.
- Developed **Spark scripts** and **UDFs** using **Spark DSL** and **Spark SQL** for data aggregation and querying, writing data back into **RDBMS** through **Sqoop**, and maintained **CI/CD** pipelines to enhance automation and deployment efficiency.
- Integrated **Azure Data Factory** with **Blob Storage** to move data through **Databricks** for processing, , enhancing data flow efficiency to **Azure Data Lake Storage** and **Azure SQL data warehouse**.
- Utilized **Jira** for ticketing and tracking issues and **Jenkins** for CI/CD, improving project management and deployment efficiency.
- Migrated data to the cloud (**Snowflake**) from legacy data warehouses and developed the infrastructure, ensuring data integrity and improving system scalability. Built **PySpark** code to validate data from raw source to **Snowflake tables**.
- Developed **Kibana** Dashboards based on the Log stash data and Integrated different source and target systems into Elasticsearch, enabling near real-time log analysis for monitoring end-to-end transactions.
- Implemented **Synapse Integration** with **Azure Databricks** notebooks which reduce about half of development work. Worked on Jenkins pipelines to run various steps including unit, integration and static analysis tools.
- Served as the subject matter expert for production system operations, overseeing the monitoring, debugging, troubleshooting, and efficient resolution of production issues.

Environment: Azure, Azure Data Lake, Blob, CI/CD, Data Factory, Elasticsearch, ETL, Factory, NoSQL, MongoDB, HDFS, Hive, Java, Jenkins, Jira, JS, Kafka, lake, Lake, MapR, Oracle, PaaS, Pig, PySpark, RDBMS, Snowflake, Spark, Spark SQL, Spark Streaming, SQL, Sqoop, Tableau, Teradata, Python

Client: L&T Technology Services (Ericsson), Vadodara, Gujarat, India

Jun 2020 - June 2022

Role: GCP Data Engineer

Description: Ericsson delivers cutting-edge telecommunications and networking solutions, driving innovation in 5G, cloud, and IoT to enhance connectivity and customer experiences worldwide. Developed and optimized data pipelines, real-time ingestion systems, and business reports using GCP, Databricks, and Apache Kafka, ensuring efficient data processing and accurate insights.

Responsibilities:

- Combined various datasets in **HIVE** to generate business reports, utilizing **Cloud Shell** for task deployment and service management. Monitored **Spark cluster** using **Log Analytics** and **Ambari Web UI**, and transitioned log storage from **Cassandra** to **Azure SQL Data Warehouse**, improving query performance.

- Created Data Studio reports to review billing and usage of services, optimizing queries and contributing to cost-saving measures. Utilized **GCP Dataproc, GCS, Cloud Functions, and BigQuery**.
- Created and provisioned different **Databricks** clusters for batch and continuous streaming data processing, installing required libraries to ensure efficient data handling.
- Developed and implemented **APIs** to automate data ingestion and transformation processes, enabling real-time data synchronization across multiple systems
- Developed and maintained a real-time data ingestion pipeline using **Java/Scala** and **Apache Kafka**, enabling near-real-time analysis of critical business metrics. Utilized **GCP Cloud Storage, Dataproc, Data Flow, BigQuery, and Glacier** for efficient data management.
- Deployed **microservices** and developed modules using **Python** and **Shell Scripting**, using **Cloud Shell SDK** to configure **GCP** services like **Dataproc, Storage, and BigQuery**.
- Loaded data into **Cassandra NoSQL** Database and integrated with **BigQuery** for large-scale data management and analysis, improving data accessibility and processing speed.
- Processed and loaded bound and unbound data from **Google Pub/Sub** topic to **BigQuery** using **Cloud Dataflow** with **Python**, and created **BigQuery** authorized views for row-level security, or exposing the data to other teams.
- Build data pipelines in **Airflow/Composer** for orchestrating **ETL** jobs using different **airflow** operators, improving workflow automation.
- Developed, deployed, and maintained **SQL Server Reporting Services (SSRS)** reports to deliver actionable insights to business stakeholders. Created parameterized, drill-down, and drill-through reports to meet complex reporting requirements, ensuring data accuracy and clarity.
- Assessed data quality and resolved inconsistencies using **SQL, Python, and Spark**, ensuring reliable and accurate datasets

Environment: Airflow, Analytics, Apache, Azure, BigQuery, Cassandra, Cluster, Data Factory, ETL, Factory, GCP, Java, Kafka, Python, Scala, SDK, Spark, SQL, VPC, NoSql, Hive

Client: Tata AIA Life Insurance, Mumbai, India

March 2019 – May 2020

Role: Data Engineer

Description: Tata AIA Life Insurance offers comprehensive life, health, and wealth management solutions, combining the trusted Tata brand with AIA Group's expertise to secure financial futures. Optimized ETL pipelines, developed data solutions on Azure and Snowflake, and automated workflows with PySpark, Kafka, and Airflow to enhance data processing efficiency.

Responsibilities:

- Developed custom multi-threaded **Java**-based ingestion jobs and **Sqoop** jobs for ingesting from FTP servers and data warehouses.
- Conducted preliminary data quality checks on the extracted data and performed end-to-end data validation, ensuring the accuracy of the KPIs.
- Developed **PySpark** applications for various **ETL** operations across data pipelines.
- Implemented **Partitioning, Dynamic Partitions, Buckets** in **Hive**, optimizing query performance and data management.
- Architected & implemented medium to large scale **BI solutions** on **Azure** using **Azure Data Platform** services (**Azure Data Lake, Data Factory, Data Lake Analytics, Stream Analytics**).
- Designed and implemented **NoSQL** database solutions **MongoDB** to store and manage large-scale, unstructured policy and claims data for an insurance client, optimizing data retrieval for real-time risk assessment and improving overall system scalability.
- Staged the **API** and **Kafka Data** (in **JSON** file format) into **Snowflake DB** by flattening it for different functional services.
- Developed Databricks **Spark** jobs with **PySpark** to perform several table operations, improving data processing efficiency.
- Integrated **HDFS** with **Apache Spark** to perform distributed data processing and analytics on vast insurance datasets. Also created data frames using **Spark SQL**.
- Designed the business requirement collection approach based on the project scope and **SDLC** methodology.
- Managed large datasets with Panda data frames and **SQL**, boosting processing efficiency.
- Developed **JSON Scripts** for deploying the Pipeline in **Azure Data Factory (ADF)** that process the data using the **Sql Activity** and creating **UNIX shell scripts** for database connectivity and executing queries in parallel job execution.
- Implemented **Apache Airflow** for workflow automation and scheduling tasks and created **DAGs tasks**.
- Migrated data from on-premises **SQL server** to **Azure Synapse Analytics** and **Azure SQL DB**.
- Developed **ETL pipelines** between data warehouses utilizing **Python, Snowflake, and SnowSQL**, including crafting SQL queries for Snowflake.
- Created and triggered automated builds and Continuous Deployments (**CI/CD**) using **Jenkins/looper** and **OneOps** cloud.

Environment: Airflow, Analytics, Apache, API, Azure, Azure Data Lake, Azure Synapse Analytics, CI/CD, Data Factory, ETL, Hive, Java, Jenkins, JS, Kafka, Kubernetes, lake, Lake, Lambda, MongoDB, PySpark, Python, Snowflake, Spark, Spark SQL, SQL, Sqoop, NoSql, HDFS

EDUCATION

- **Masters in Information and technology management, USA**